

Business Report: Online Retail Customer Sales Analysis

Executive Summary

This project analyzes customer transactions from an online retail store between December 2009 and December 2011. The objective was to identify sales trends, understand customer behavior, evaluate product performance, and generate actionable business recommendations.

Using Python for data analysis and Power BI for visualization, the project uncovered key insights related to revenue growth, customer segmentation, and product contribution.

Project Objectives

The main objectives of this analysis were:

- Analyze overall sales performance
 - Identify revenue trends over time
 - Understand customer purchasing behavior
 - Segment customers using RFM Analysis
 - Evaluate product performance
 - Generate business recommendations for growth
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Dataset Overview

The dataset contains transactional information including:

- Invoice Number
- Product Information
- Quantity Purchased
- Unit Price
- Customer ID
- Country
- Transaction Date

After cleaning and preprocessing, the dataset was prepared for exploratory and business analysis.

Key Findings

1. Sales Performance

Observation

- Total Revenue exceeded £20 million.
- More than 40,000 orders were recorded.
- Approximately 6,000 customers made purchases.

Insight

The business demonstrates strong sales volume and a broad customer base, indicating a healthy e-commerce operation.

2. Revenue Trend Analysis

Observation

- Revenue generally increased throughout the period.
- Significant growth occurred during the final quarter of the year.
- November recorded the highest monthly revenue.

Insight

Seasonal demand strongly influences sales performance, particularly during holiday shopping periods.

Recommendation

- Increase inventory before Q4.
 - Launch promotional campaigns during peak shopping seasons.
 - Prepare logistics and fulfillment resources for year-end demand.
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3. Geographic Performance

Observation

- The United Kingdom generated the majority of total revenue.

- Other countries contributed significantly smaller revenue shares.

Insight

The business is highly dependent on the UK market.

Recommendation

- Expand marketing efforts in high-potential international markets.
 - Explore localized promotions and shipping incentives.
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4. Customer Segmentation Analysis

RFM (Recency, Frequency, Monetary) analysis was used to classify customers.

Segment Distribution

Segment	Description
Champions	High-value and highly engaged customers
Loyal Customers	Frequent repeat customers
Potential Loyalists	Customers showing growth potential
At Risk	Previously active customers becoming inactive
Others	Remaining customer groups

Observation

- Champions generated the highest revenue contribution.
- Loyal Customers formed the second-largest valuable segment.
- At Risk customers represented a notable portion of the customer base.

Insight

A relatively small group of customers drives a large percentage of overall revenue.

Recommendation

- Reward Champions through loyalty programs.
 - Encourage repeat purchases among Loyal Customers.
 - Re-engage At Risk customers using targeted email campaigns and discounts.
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5. Product Performance Analysis

Observation

- A limited number of products generated a large share of total revenue.
- Product demand was concentrated among top-selling items.

Insight

The business follows a Pareto-like pattern where a small percentage of products contributes disproportionately to revenue.

Recommendation

- Maintain strong inventory levels for top-performing products.
 - Use high-performing products in promotional campaigns.
 - Review underperforming products for optimization or removal.
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Business Impact

The analysis highlights several opportunities:

1. Increase customer retention through targeted marketing.
 2. Focus on high-value customer segments.
 3. Optimize inventory planning using demand trends.
 4. Expand internationally to reduce dependency on a single market.
 5. Improve profitability through product portfolio optimization.
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Conclusion

The Online Retail Customer Sales Analysis project successfully identified critical sales, customer, and product insights. Revenue growth is driven primarily by a loyal customer base and a small group of top-performing products. By implementing customer retention strategies, improving inventory planning, and expanding market reach, the business can further increase revenue and long-term customer value.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- Jupyter Notebook